# 20 Easy Image Processing Programs

Very simple Python code for Jupyter Notebook exam practice

## Setup (run once in Jupyter):

```
%pip install opencv-python numpy matplotlib
```

**Important:** Put your image in the same folder as the notebook. Replace image.jpg with your filename. For Question 17 use document.jpg; for Question 19 use image1.jpg and image2.jpg. Matplotlib is used to show the output in Jupyter.

## 1. Grayscale

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

plt.imshow(gray, cmap="gray")
plt.show()
```

## 2. Resize Image

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg")
small = cv2.resize(img, (200, 200))

plt.imshow(cv2.cvtColor(small, cv2.COLOR_BGR2RGB))
plt.show()
```

## 3. Rotate Image

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg")
rotated = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)

plt.imshow(cv2.cvtColor(rotated, cv2.COLOR_BGR2RGB))
plt.show()
```

## 4. Increase Brightness

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg")
bright = cv2.convertScaleAbs(img, beta=50)

plt.imshow(cv2.cvtColor(bright, cv2.COLOR_BGR2RGB))
plt.show()
```

## 5. Increase Contrast

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg")
contrast = cv2.convertScaleAbs(img, alpha=2)

plt.imshow(cv2.cvtColor(contrast, cv2.COLOR_BGR2RGB))
plt.show()
```

## 6. Gaussian Blur

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg")
blur = cv2.GaussianBlur(img, (5, 5), 0)

plt.imshow(cv2.cvtColor(blur, cv2.COLOR_BGR2RGB))
plt.show()
```

## 7. Edge Detection

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg", 0)
edges = cv2.Canny(img, 100, 200)

plt.imshow(edges, cmap="gray")
plt.show()
```

## 8. Thresholding

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg", 0)
_, binary = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)

plt.imshow(binary, cmap="gray")
plt.show()
```

## 9. Crop Image

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg")
crop = img[50:200, 50:200]

plt.imshow(cv2.cvtColor(crop, cv2.COLOR_BGR2RGB))
plt.show()
```

## 10. Detect Red Color

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg")
hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)

low = np.array([0, 100, 100])
high = np.array([10, 255, 255])
mask = cv2.inRange(hsv, low, high)
result = cv2.bitwise_and(img, img, mask=mask)

plt.imshow(cv2.cvtColor(result, cv2.COLOR_BGR2RGB))
plt.show()
```

## 11. Histogram

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg", 0)
plt.hist(img.ravel(), 256, [0, 256])
plt.show()
```

## 12. Histogram Equalization

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg", 0)
equal = cv2.equalizeHist(img)

plt.imshow(equal, cmap="gray")
plt.show()
```

## 13. Salt-and-Pepper Noise + Median Filter

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg", 0)
noisy = img.copy()
noisy[::10, ::10] = 0
noisy[5::10, 5::10] = 255
clean = cv2.medianBlur(noisy, 3)

plt.imshow(clean, cmap="gray")
plt.show()
```

## 14. Sharpen Image

```
import cv2
import matplotlib.pyplot as plt
import numpy as np

img = cv2.imread("image.jpg")
kernel = np.array([[0, -1, 0], [-1, 5, -1], [0, -1, 0]])
sharp = cv2.filter2D(img, -1, kernel)

plt.imshow(cv2.cvtColor(sharp, cv2.COLOR_BGR2RGB))
plt.show()
```

## 15. Find Contours

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg", 0)
_, binary = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)
contours, _ = cv2.findContours(binary, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
result = cv2.cvtColor(img, cv2.COLOR_GRAY2BGR)
cv2.drawContours(result, contours, -1, (0, 255, 0), 2)

plt.imshow(cv2.cvtColor(result, cv2.COLOR_BGR2RGB))
plt.show()
```

## 16. Erosion and Dilation

```
import cv2
import matplotlib.pyplot as plt
import numpy as np

img = cv2.imread("image.jpg", 0)
_, binary = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)
kernel = np.ones((3, 3), np.uint8)

erode = cv2.erode(binary, kernel)
dilate = cv2.dilate(binary, kernel)

plt.imshow(dilate, cmap="gray")
plt.show()
```

## 17. Clean Document Image

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("document.jpg", 0)
clean = cv2.adaptiveThreshold(img, 255, cv2.ADAPTIVE_THRESH_GAUSSIAN_C,
                              cv2.THRESH_BINARY, 11, 2)

plt.imshow(clean, cmap="gray")
plt.show()
```

## 18. Face Detection

```
import cv2
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
face = cv2.CascadeClassifier(cv2.data.haarcascades + "haarcascade_frontalface_default.xml")
faces = face.detectMultiScale(gray, 1.1, 5)

for x, y, w, h in faces:
    cv2.rectangle(img, (x, y), (x+w, y+h), (0, 255, 0), 2)

plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.show()
```

## 19. Image Difference

```
import cv2
import matplotlib.pyplot as plt

img1 = cv2.imread("image1.jpg")
img2 = cv2.imread("image2.jpg")
img2 = cv2.resize(img2, (img1.shape[1], img1.shape[0]))
diff = cv2.absdiff(img1, img2)

plt.imshow(cv2.cvtColor(diff, cv2.COLOR_BGR2RGB))
plt.show()
```

## 20. Green Color Segmentation

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread("image.jpg")
hsv = cv2.cvtColor(img, cv2.COLOR_BGR2HSV)
low = np.array([25, 40, 40])
high = np.array([90, 255, 255])
mask = cv2.inRange(hsv, low, high)
leaf = cv2.bitwise_and(img, img, mask=mask)

plt.imshow(cv2.cvtColor(leaf, cv2.COLOR_BGR2RGB))
plt.show()
```